Study of Feature Extraction Methods for Motor Imagery (MI) EEG Classification

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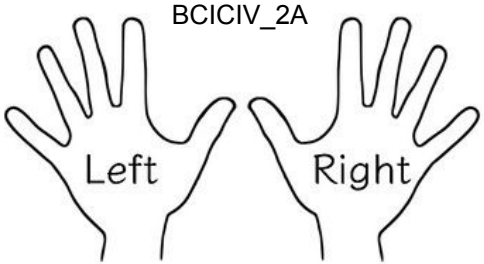
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Outlines

- Motivation
- Method Rationale
- Leakage prevention
- Results
- Interpretation
- Conclusion

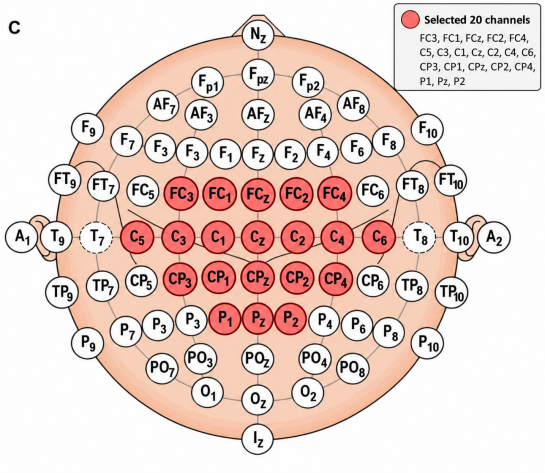
Motivations

- First, we are interested in band power and time-frequency domain (STFT) of EEG analyzation to compare with FBCSP of MI task .
- Next, we designed 2 lightweight models: DNN and Conv2D-based model suited for band power features and STFT respectively.
- Then, we found Common Average Reference and Differential asymmetry [1] improving the performance of using STFT.
- Finally, we proposed the STFT features and the lightweight model based CNN2D to classify MI task (Left hand and Right hand)



Methods

EEG Montage and Motor Imagery (MI)



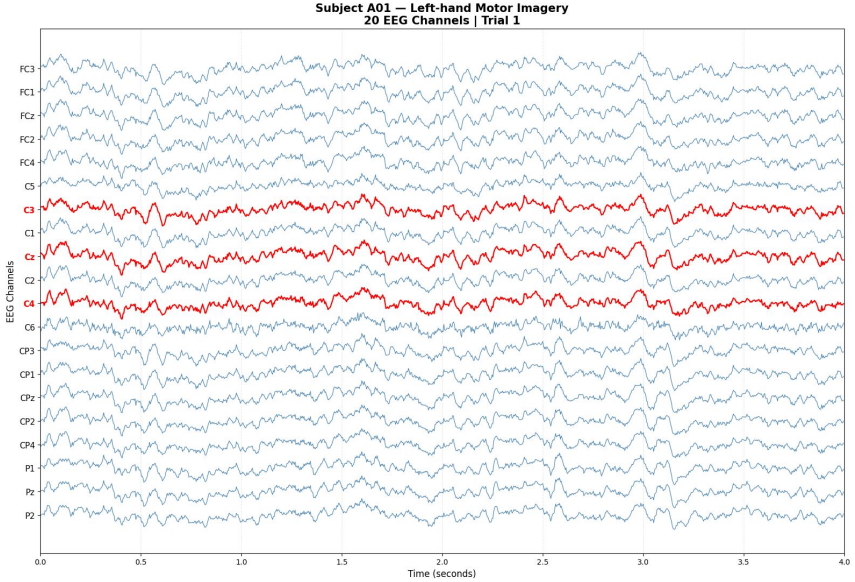
Referential montage

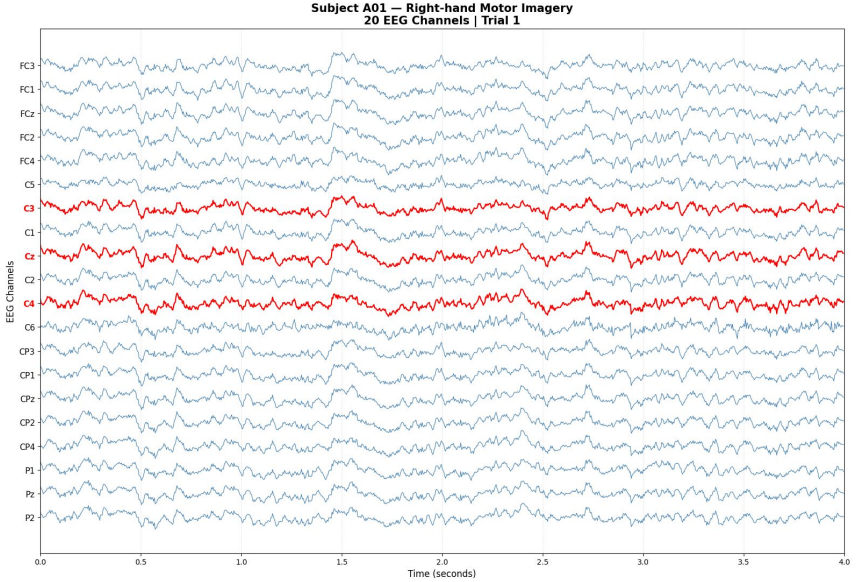
- Motor Imagery (MI): Imagining movement without physical execution.
- EEG montage: Electrodes positioned across the scalp to record brain activity.

https://link.springer.com/chapter/10.1007/978-3-642-38256-7_18

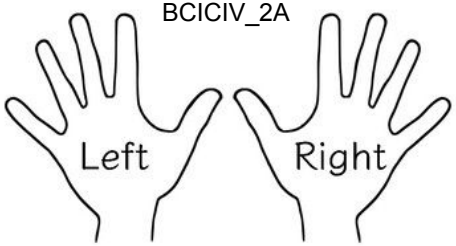
Methods

20-Channel EEG Plots for Left-Hand and Right-Hand Motor Imagery





BCICIV_2A



Methods

Common Average Reference (CAR)

For an array of N electrodes, each recording a voltage at time t, written as $V_1(t)$, $V_2(t)$, up through $V_N(t)$, the CAR-transformed value for any single electrode i is:

$$V_i^{CAR}(t) = V_i(t) - \frac{V_1(t) + V_2(t) + \cdots + V_N(t)}{N}$$

- This helps reduce signal components shared across electrodes and emphasizes spatial differences in brain activity.
- Importantly, CAR preserves the original dimensions of the EEG signals.
- After re-referencing, the EEG signals are ready for frequency-based analysis.

https://www.emotiv.com/neuroscience/common-average-reference-eeg?srsltid=AU7gw4WJxV_64OLYzhwngVx2fLqgFWwUy-EPxtLNs1bs6AT7F3EO7jxY

Methods

Band Power

- Measures EEG power in different frequency bands.
  - Theta: 4-8 Hz
  - Alpha: 8-13 Hz
  - Beta: 13-30 Hz
  - Gamma: 30-40 Hz

- Estimates PSD (Power Spectral Density) using Welch’s method and integrates it over each frequency band to obtain band power.

Methods

Short time Fourier Transform (STFT)

- STFT is a special flavor of a Fourier transform where you can see how your frequencies in your signal change through time.
- STFT has several parameters to tune, including nperseg (window length), noverlap (overlap between windows), nfft (FFT size), and win_s (window duration in seconds). These parameters affect time-frequency resolution, computational cost, and classification performance.

Problem with Short time Fourier Transform (STFT)

- CSP works well because it combines spatial filtering with log-variance features. We therefore aimed to keep the interpretable STFT representation and address that limitations by applying Differential Asymmetry (DASM).

Methods

Differential Asymmetry (DASM)

- DASM calculates the difference in Differential Entropy (DE) between pairs of EEG electrodes in the left and right hemispheres (FC3–FC4, FC1–FC2, C5–C6, C3–C4, C1–C2, CP3–CP4, CP1–CP2, and P1–P2).
- Differential Entropy (DE) measures the information content of a continuous EEG signal and is commonly used as a feature in different frequency bands. [1]

DE was defined as

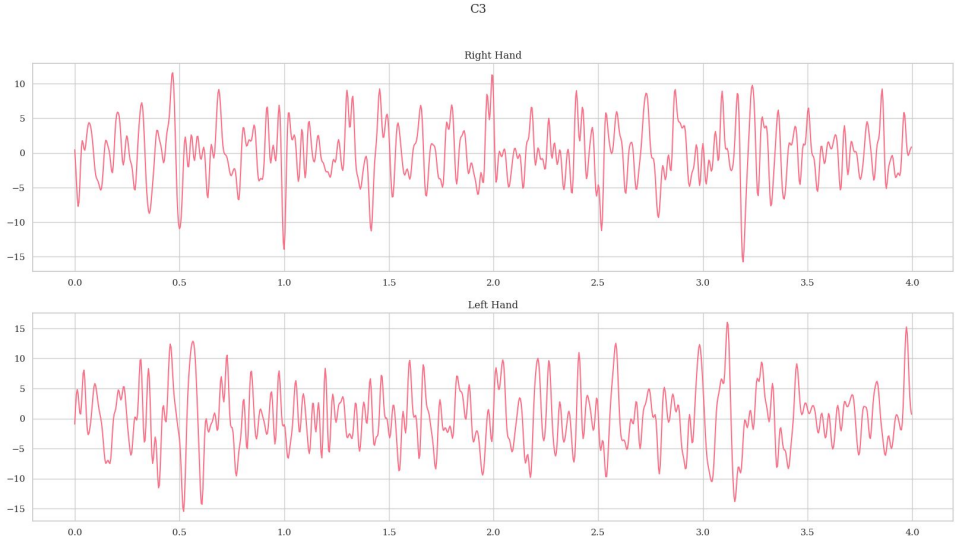
$$h(X) = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \log\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right) dx = \frac{1}{2}\log(2\pi e\sigma^2), \quad (1)$$

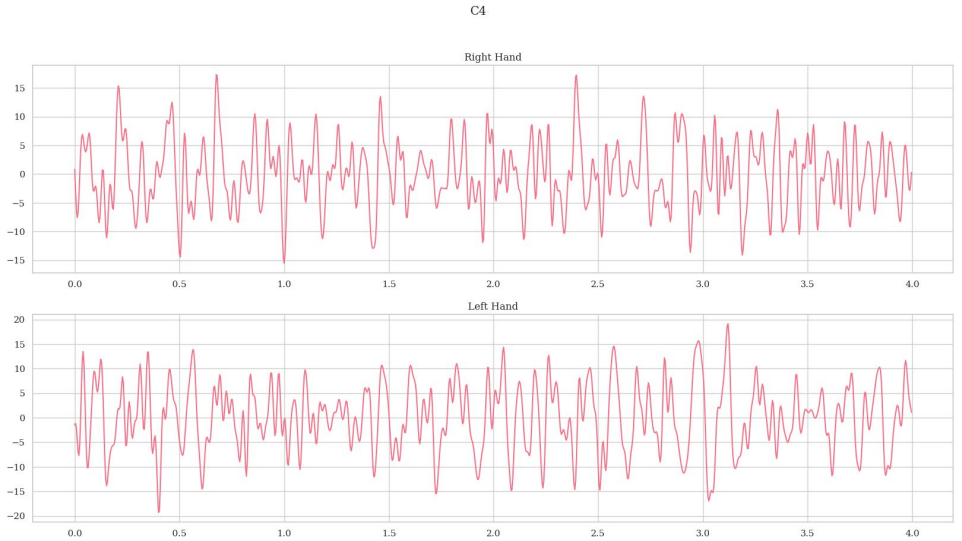
where the time series X obeys the Gauss distribution $N(\mu, \sigma^2)$. It has been proven that, for a fixed length EEG sequence, DE is equivalent to the logarithm ES in a certain frequency band [7]. DE was employed to construct features in five frequency bands mentioned above.

$$DASM = h(X_i^{left}) - h(X_i^{right}) \quad (2)$$

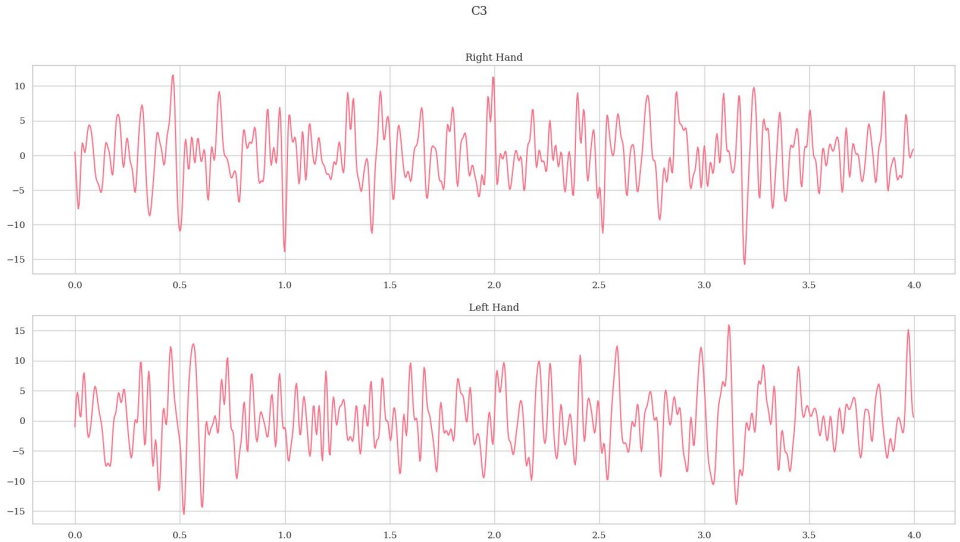
where h(X) is defined in Eq. 1, and i is the pair number.

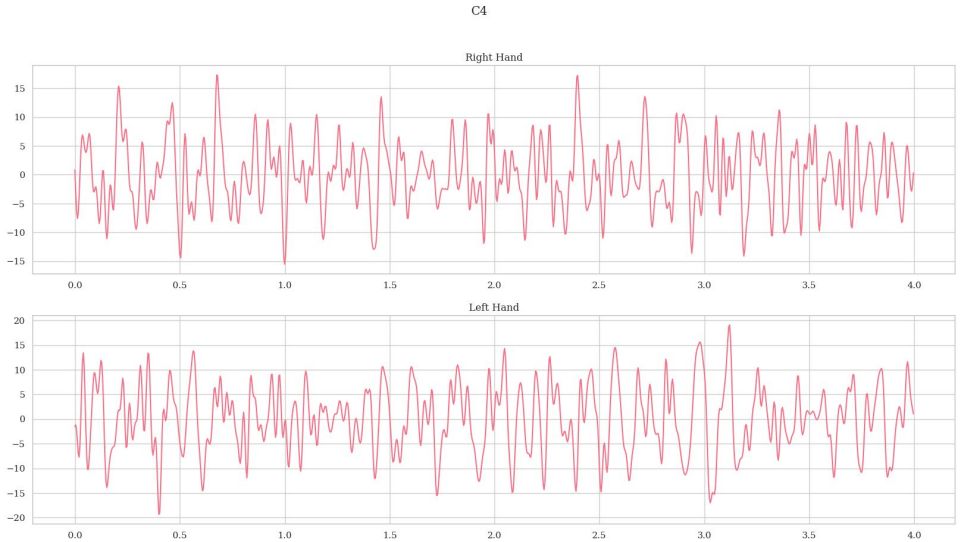
Raw EEG with Bandpass filter



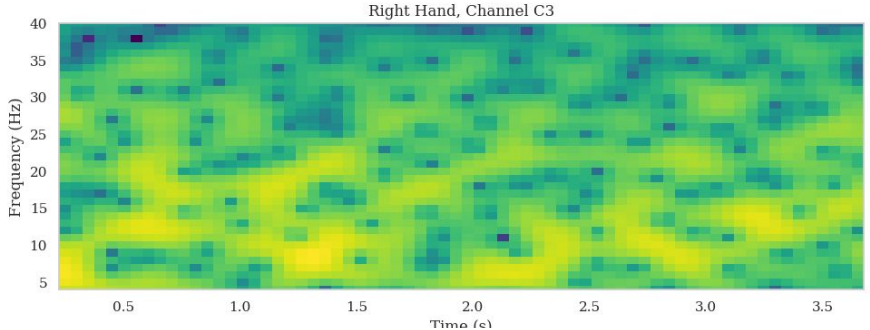


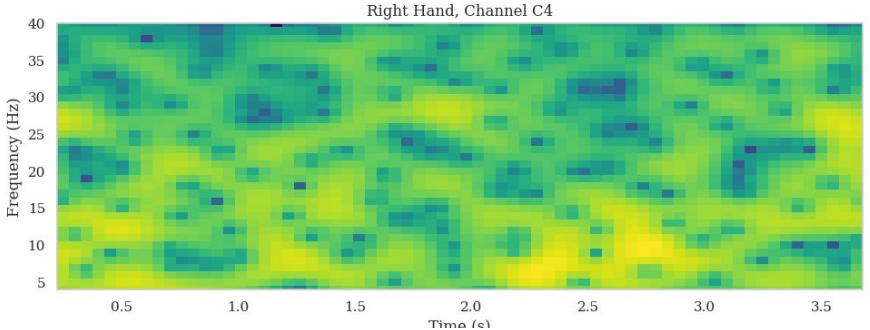
Raw EEG + BP + CAR

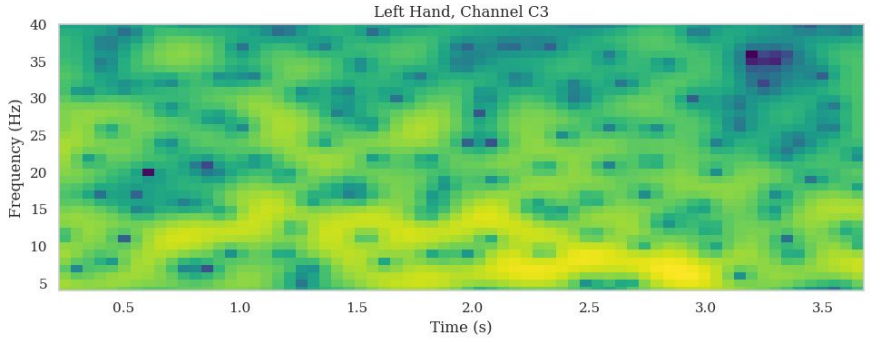


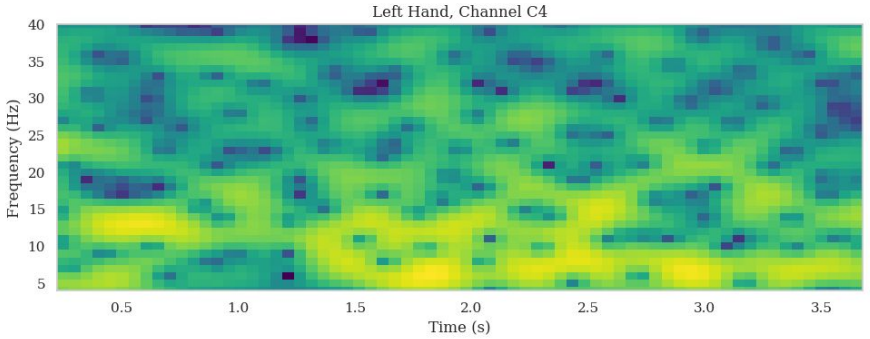


Raw EEG + BP + CAR + STFT

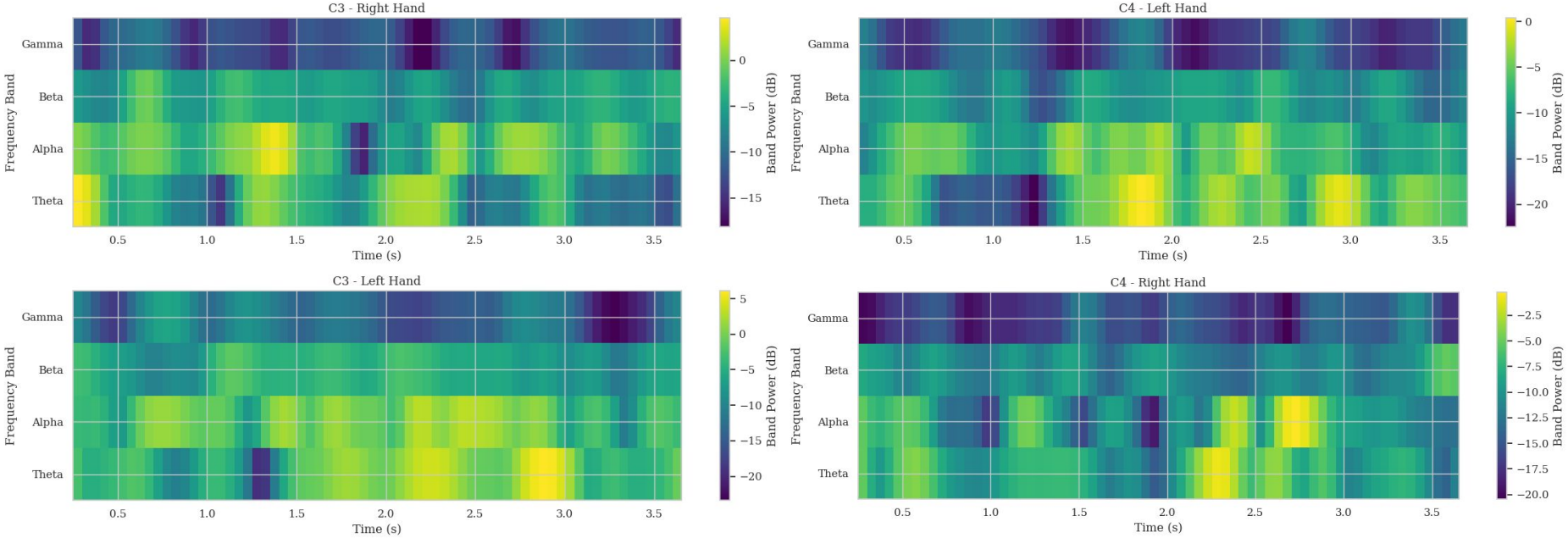




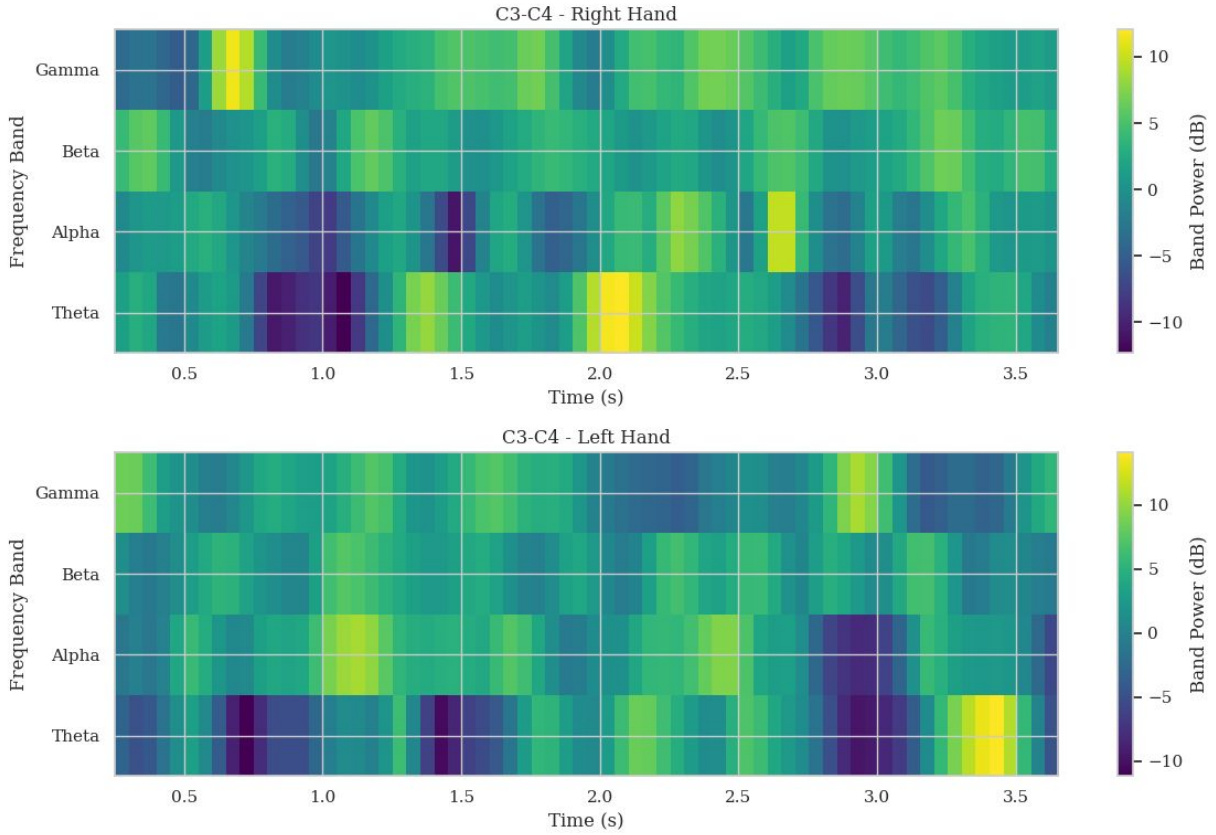




Raw EEG + BP + CAR + STFT + Filtering Bands



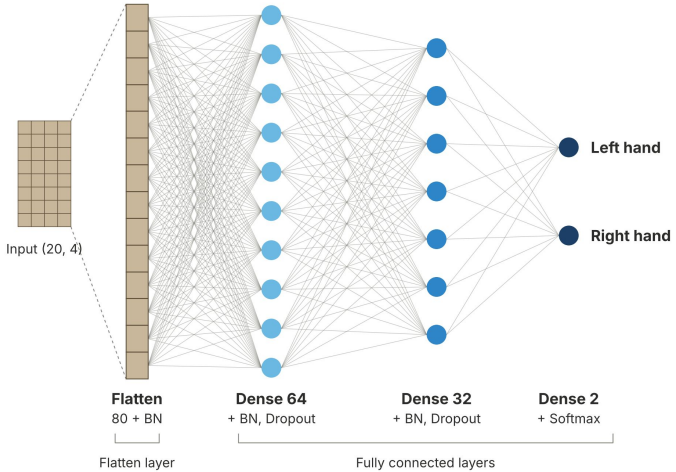
Raw EEG + BP + CAR + STFT + Filtering Bands + DASM



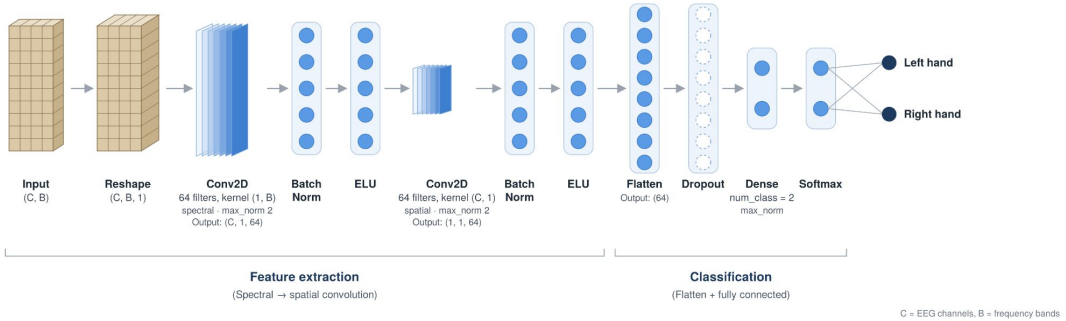
Methods

Model Architectures

1. Deep Neural Network for Band Power Features



2. Convolutional Neural Network for STFT features



Methods

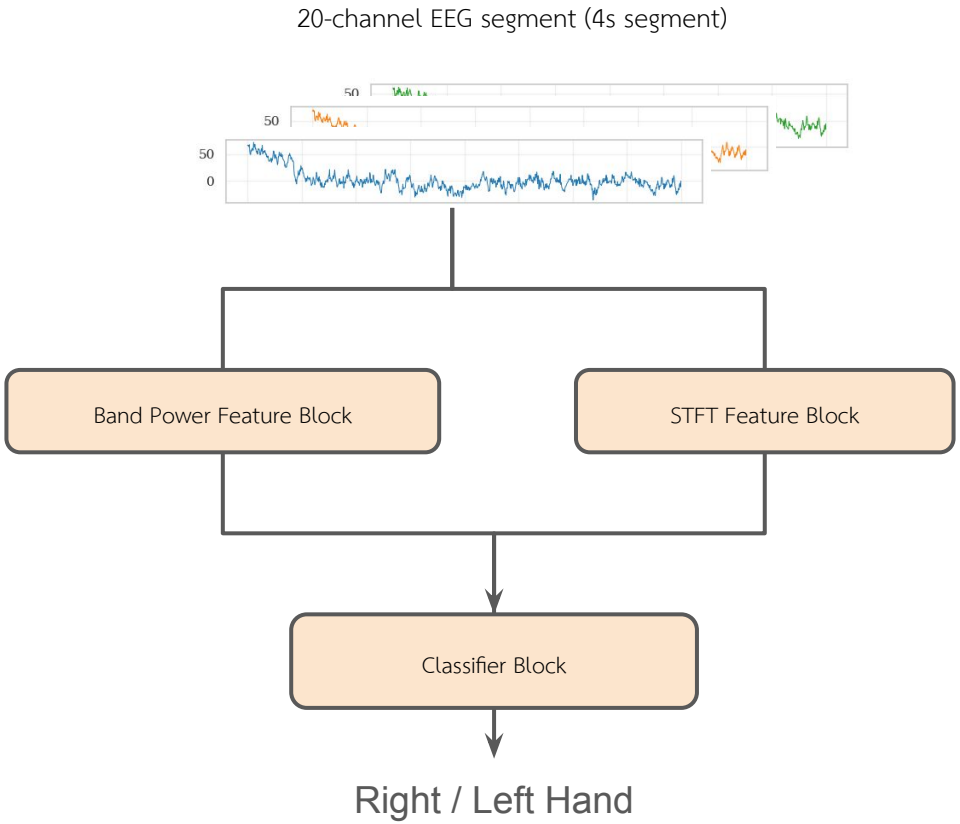
Hyperparameters

- **Optimizer:** Adam (learning rate = 0.001)
- **Loss function:** Sparse Categorical Cross Entropy
- **Batch size:** 16
- **Maximum epochs:** 200
- **Dropout:** 0.5
- **Early stopping:** Patience of 20 epochs
- **Learning rate scheduler:** Reduce by a factor of 0.5 after 5 epochs without improvement
- **Class balancing:** Calculate Class Weights

Leakage Prevention

- Our experiment used a subject-dependent setting. We applied stratified 5-fold cross-validation to the segments from the same subject in Session T and evaluated the model on the test data from Session E.

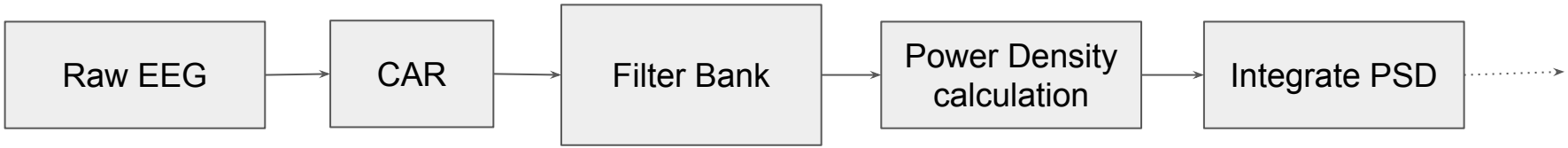
Overview


20-channel EEG segment (4s segment)
50
50
50
0
Band Power Feature Block
STFT Feature Block
Classifier Block
Right / Left Hand

Pipeline

- 20-channel EEG segments (4 seconds each)
- **Filter bank :** Theta, Alpha, Beta and Gamma
- **Feature extraction**
  - Band power Feature
  - STFT (time–frequency representation) Features
- **Normalize:** Standardize all Features by channels
- **Model:** Lightweight neural network
- **Output:** Two classes — left-hand and right-hand motor imagery

Overview of Band Power Features

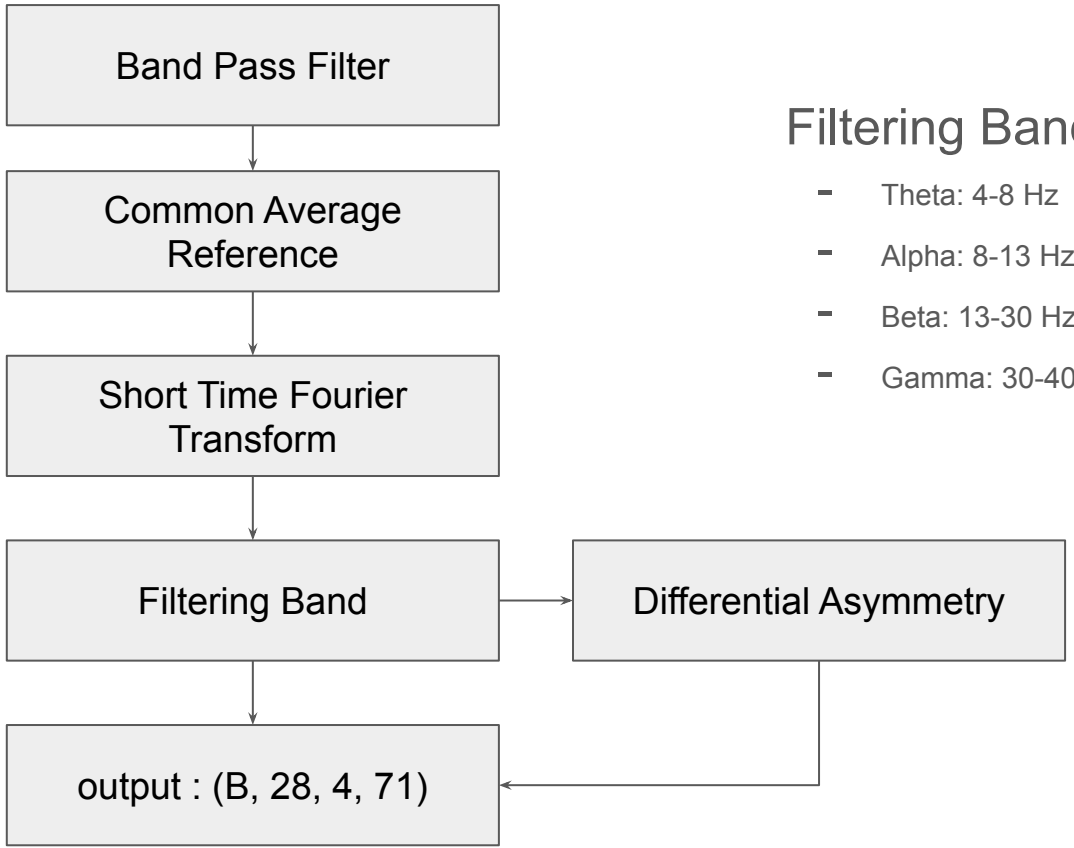


Theta: 4-8 Hz
Alpha: 8-13 Hz
Beta: 13-30 Hz
Gamma: 30-40 Hz

Shape : (...,20,4)

(N_sample, N_channels, N_features)

Overview of STFT Feature



Filtering Band

- Theta: 4-8 Hz
- Alpha: 8-13 Hz
- Beta: 13-30 Hz
- Gamma: 30-40 Hz

Results

Feature	Model	Accuracy (%)	Macro F1-score (%)
FBCSP	**MixNet**	**76.48 ± 15.17%**	**75.23 ± 16.44%**
Band power features without norm per channel	DNN	61.14 ± 11.83	56.71 ± 14.32
Band power features with norm per channel	DNN	62.99 ± 12.01	61.12 ± 13.55
CAR + Band power features with norm per channel	DNN	63.09 ± 12.01	61.22 ± 13.55
CAR->STFT->Filtering Band	CNN2D	72.70 ± 13.90	71.90 ± 14.60
CAR->STFT->Filtering Band->concat DASM	**CNN2D**	**75.12 ± 13.06%**	**74.67 ± 13.49%**

Mean ± S.D.

Ablation Study of STFT Parameters

Preprocess	STFT parameters	Acc	Macro F1
Raw->CAR->STFT->Filtering Band	Nfft = 250, win_s = 0.5, overlap 90%	0.727 ± 0.139	0.719 ± 0.146
Raw->CAR->STFT->Filtering Band->DASM	Nfft 250, win_s = 0.5, overlap 90%	0.721 ± 0.128	0.717 ± 0.131
Raw->CAR->STFT->Filtering Band->concat DASM	Nfft = 250, win_s=1, overlap 50%	0.737 ± 0.136	0.731 ± 0.142
Raw->CAR->STFT->Filtering Band->concat DASM	Nfft = 250, win_s=0.5, overlap 50%	0.741 ± 0.134	0.7332 ± 0.1417
Raw->CAR->STFT->Filtering Band->concat DASM	Nfft = 250, win_s = 0.5, overlap 90%	0.7437 ± 0.1337	0.7384 ± 0.1388

Mean ± S.D.

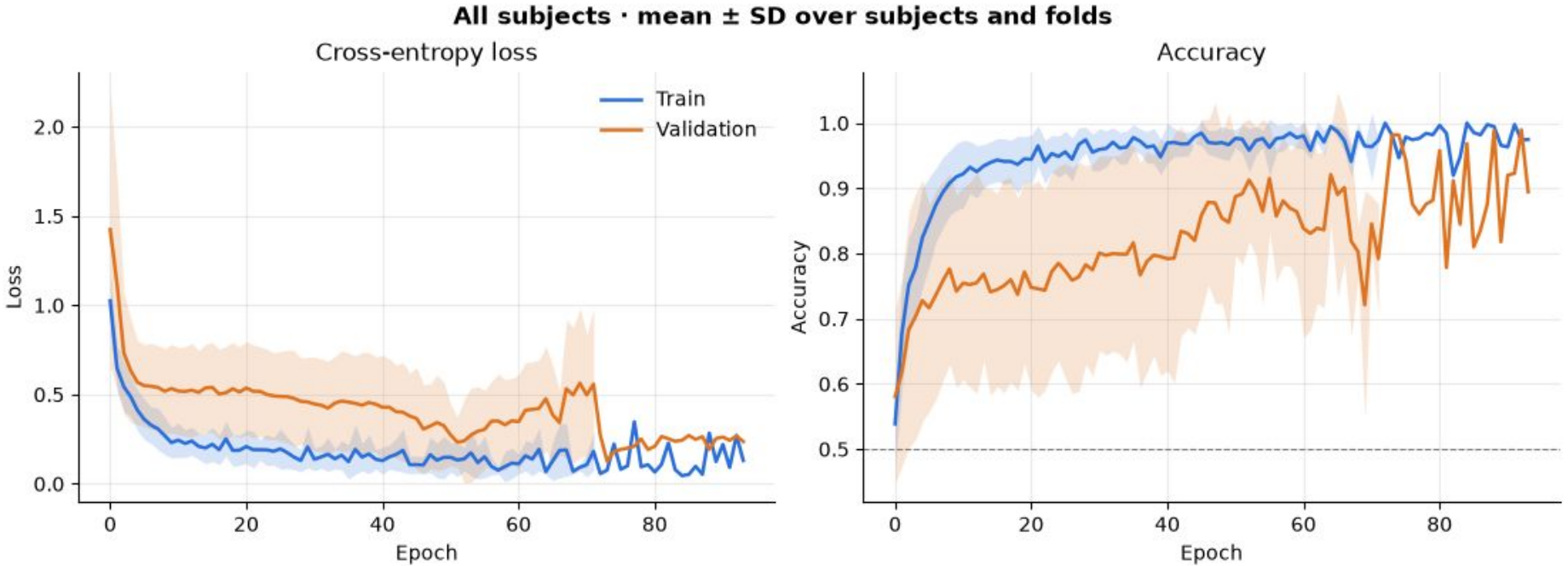
Ablation Study of Filter Size and Dropout in CNNs

(Raw->CAR->STFT->Filtering Band->concat DASM)

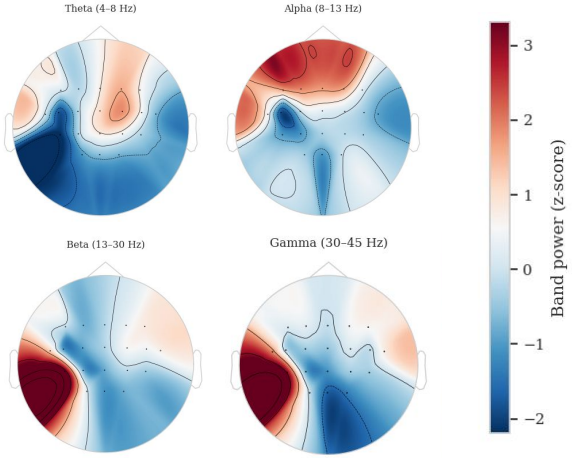
Filter Size	Dropout	Accuracy (%)	F1-score (%)
32	0.3	74.55 ± 13.63	74.01 ± 14.16
32	0.4	74.72 ± 13.60	74.19 ± 14.17
64	0.3	74.60 ± 13.27	74.10 ± 13.76
64	**0.4**	**75.12 ± 13.06**	**74.67 ± 13.49**
64	0.5	74.49 ± 13.73	73.93 ± 14.29
64	0.6	74.63 ± 13.54	74.13 ± 14.01
128	0.4	73.95 ± 13.84	73.39 ± 14.37
128	0.5	74.52 ± 13.41	73.97 ± 13.95

Mean ± S.D.

Loss (Raw -> CAR -> STFT -> Filtering Band -> concat DASM)



Interpretation



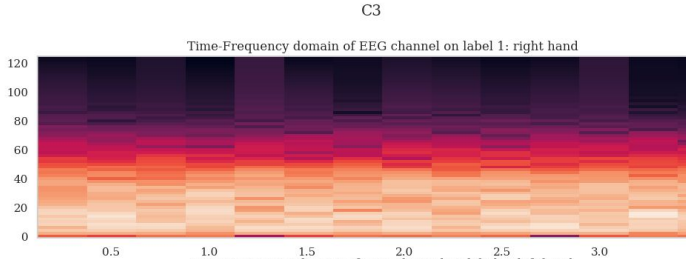
Band Power

Advantages

- Simple and computationally efficient.
- Captures power in different frequency bands.
- Suitable for small datasets.

Limitations

- Loses temporal information when power is averaged over a segment.
- May not capture detailed time-frequency patterns.



STFT Differential Asymmetry Maps

Advantages

- Captures time-frequency patterns.
- Represents differences between left and right EEG channels.
- Can provide useful spatial information for motor imagery classification.

Limitations

- Requires more computation and memory.
- Depends on STFT parameters and channel pairing.
- May lose common activity shared by both hemispheres.

Conclusion

- The proposed method achieved an accuracy of 75.12 ± 13.06%. CAR helped reduce common noise in EEG signals, while DASM captured differences between the left and right hemispheres. These techniques improved the quality of EEG features for motor imagery classification.

References

[1]. R. -N. Duan, J. -Y. Zhu and B. -L. Lu, "Differential entropy feature for EEG-based emotion classification," 2013 6th International IEEE/EMBS Conference on Neural Engineering (NER), San Diego, CA, USA, 2013, pp. 81-84, doi: 10.1109/NER.2013.6695876.